

Generative artificial intelligence (Generative AI or GenAI) is a subfield of [artificial intelligence](#) that uses [generative models](#) to generate text, [images](#), [videos](#), [audio](#), [software code](#) or other forms of data.^[1] These models [learn](#) the underlying patterns and structures of their [training data](#) and use them to generate new data^[2] in response to input, which often takes the form of natural language [prompts](#).^{[3][4]}

The prevalence of generative AI tools has increased significantly since the [AI boom](#) in the 2020s. This boom was made possible by improvements in [deep neural networks](#), particularly [large language models](#) (LLMs), which are based on the [transformer](#) architecture. Generative AI applications include [chatbots](#) such as [ChatGPT](#), [Claude](#), [Copilot](#), [DeepSeek](#), [Google Gemini](#) and [Grok](#); [text-to-image](#) models such as [Stable Diffusion](#), [Midjourney](#), and [DALL-E](#); and [text-to-video](#) models such as [Veo](#), [LTX](#) and [Sora](#).^{[5][6][7]} Technology companies developing generative AI include [Alibaba](#), [Anthropic](#), [Baidu](#), [DeepSeek](#), [Google](#), [Lightricks](#),^[8] [Meta AI](#), [Microsoft](#), [Mistral AI](#), [OpenAI](#), [Perplexity AI](#), [xAI](#),^[9] and [Yandex](#).^[10]

Companies in a variety of sectors have used generative AI, including those in software development, healthcare,^[11] finance,^[12] entertainment,^[13] customer service,^[14] sales and marketing,^[15] art, writing,^[16] and product design.^[17]

Applications

Notable types of generative AI models include [generative pre-trained transformers](#) (GPTs), [generative adversarial networks](#) (GANs), and [variational autoencoders](#) (VAEs). Generative AI systems are [multimodal](#) if they can process multiple types of inputs or generate multiple types of outputs.^{[49][unreliable source?]} For example, [GPT-4o](#) can both process and generate text, images and audio.^[50]

Generative AI has made its appearance in a wide variety of industries, radically changing the dynamics of content creation, analysis, and delivery. In healthcare,^[51] for instance, generative AI accelerates [drug discovery](#) by creating molecular structures with target characteristics^[52] and generates [radiology](#) images for training diagnostic models. This ability not only enables faster and cheaper development but also enhances medical decision-making. In finance, generative AI services help create datasets and automate reports using natural language. It automates content creation, produces synthetic financial data, and tailors customer communications. It also powers chatbots and virtual agents. Collectively, these technologies enhance efficiency, reduce [operational costs](#), and support data-driven decision-making in financial institutions.^{[53][unreliable source?]} The media industry makes use of generative

AI for numerous creative activities such as music composition, scriptwriting, video editing, and digital art. The educational sector is impacted as well, since the tools make learning personalized through creating quizzes, study aids, and essay composition. Both the teachers and the learners benefit from AI-based platforms that suit various learning patterns.^[54] In the educational field, in [Colombia](#), student use of [Meta](#)'s generative AI programs resulted in a decline in scores.^[55]

Text and software code

[Large language models](#) (LLM) are trained on [tokenized](#) text from [text corpora](#). Such systems include [ChatGPT](#), [Gemini](#), [Claude](#), [LLaMA](#), and [BLOOM](#). LLMs are capable of [natural language processing](#), [machine translation](#), and [natural language generation](#).^[57] LLMs can be used as [foundation models](#) for other tasks.^[58] They can be trained on [computer code](#), which makes it possible to generate [source code](#) for new [computer programs](#) with [prompts](#), a practice known as [vibe coding](#).^[59] Examples include [OpenAI Codex](#), [Tabnine](#), [GitHub Copilot](#), [Microsoft Copilot](#), and the [VS Code fork Cursor](#).^[60] Some AI assistants help candidates cheat during online [coding interviews](#) by providing code, improvements, and explanations. Their clandestine interfaces minimize the need for eye movements that would expose cheating to the interviewer.^[61]

Images

Generative AI can be used to create [visual art](#).^[71] Such systems are trained on sets of images along with their [text captions](#). Examples of [text-to-image](#) models include [Stable Diffusion](#), [DALL-E](#), [Midjourney](#), [Imagen](#), [Recraft](#), [Adobe Firefly](#), and [Flux](#). They can also be used for [neural style transfer](#).^[72]

Video

Generative AI can be used to generate [photorealistic](#) videos. Examples include [Sora](#) by [OpenAI](#),^[7] [Runway](#),^[73] [Make-A-Video](#) by [Meta Platforms](#) and the open source LTX Video by [Lightricks](#).^[74]

Robotics

By training on [robotic](#) system motions, generative AI can create new trajectories for [motion planning](#) and [robot navigation](#).^[75] Multimodal [vision-language-action models](#) such as Google's RT-2 can perform rudimentary reasoning in response to user prompts and visual input, such as picking up a toy [dinosaur](#) when given the prompt *pick up the extinct animal* at a table filled with toy animals and other objects.^[76]

3D modeling

Artificially intelligent [computer-aided design](#) (CAD) can use text-to-3D, image-to-3D, and video-to-3D to [automate 3D modeling](#).^[77] AI-based [CAD libraries](#) could also be developed using [linked open data](#) of [schematics](#) and [diagrams](#).^[78] AI CAD [assistants](#) are used as tools to help streamline workflow.^[79]

Variational autoencoders

[Variational autoencoders](#) (VAEs) are deep learning models that probabilistically encode data. They are typically used for tasks such as [noise reduction](#) from images, [data compression](#), identifying unusual patterns, and [facial recognition](#). Unlike [standard autoencoders](#), which compress input data into a fixed latent representation, VAEs model the [latent space](#) as a probability distribution,^[98] allowing for smooth sampling and interpolation between data points. The encoder ("recognition model") maps input data to a latent space, producing means and variances that define a probability distribution. The decoder ("generative model") samples from this latent distribution and attempts to reconstruct the original input. VAEs optimize a loss function that includes both the reconstruction error and a [Kullback–Leibler divergence](#) term, which ensures the latent space follows a known prior distribution. VAEs are particularly suitable for tasks that require structured but smooth latent spaces, although they may create blurrier images than GANs. They are used for applications like image generation, data interpolation and [anomaly detection](#).

Transformers

Transformers became the foundation for many powerful generative models, most notably the [generative pre-trained transformer](#) (GPT) series developed by OpenAI. They marked a major shift in natural language processing by replacing traditional [recurrent](#) and [convolutional](#) models.^[99]^[*unreliable source?*] This architecture allows models to process entire sequences simultaneously and capture long-range dependencies more efficiently. The [self-attention mechanism](#) enables the model to capture the significance of every word in a sequence when predicting the subsequent word, thus improving its contextual understanding. Unlike recurrent neural networks, transformers process all the tokens in parallel, which improves the training efficiency and scalability. Transformers are typically pre-trained on enormous corpora in a [self-supervised](#) manner, prior to being [fine-tuned](#).^[*citation needed*]

Concerns

See also: [Ethics of artificial intelligence](#) and [Artificial intelligence controversies](#)

The development of generative AI has raised concerns from [governments](#), businesses, and individuals, resulting in protests, legal actions, calls to [pause AI experiments](#), and actions by multiple governments. In a July 2023 briefing of the [United Nations Security Council](#), [Secretary-General António Guterres](#) stated "Generative AI has enormous potential for good and evil at scale", that AI may "turbocharge global development" and contribute between \$10 and \$15 trillion to the global economy by 2030, but that its malicious use "could cause horrific levels of death and destruction, widespread trauma, and deep psychological damage on an unimaginable scale".^[118] In addition, generative AI has a significant [carbon footprint](#).^[119]

Academic honesty

Generative AI can be used to generate and modify academic prose, to paraphrase sources, and translate languages. The use of generative AI in a classroom setting can be a form of [academic plagiarism](#). Some schools have banned ChatGPT and similar tools.^{[120][121][[better source needed](#)]}

A commonly proposed use for teachers is grading and giving feedback. Companies like Pearson and ETS use AI to score grammar, mechanics, usage, and style, but not for main ideas or overall structure.^[120] The [National Council of Teachers of English](#) stated that machine scoring makes students feel their writing is not worth reading.^{[122][[non-primary source needed](#)]} AI scoring has also given unfair results for students from different ethnic backgrounds.^[123]

Racial and gender bias

Generative AI models can reflect and amplify any [cultural bias](#) present in the underlying data. For example, a language model might assume that doctors and judges are male, and that secretaries or nurses are female, if those biases are common in the training data.^[133] Similarly, an image model prompted with the text "a photo of a CEO" might disproportionately generate images of white male CEOs,^{[[citation needed](#)]} if trained on a racially biased data set. A number of methods for mitigating bias have been attempted, such as altering input prompts^[134] and reweighting training data.^[135]

Deepfakes

Deepfakes (a [portmanteau](#) of "deep learning" and "fake"^[136]) are AI-generated media that take a person in an existing image or video and replace them with someone else's likeness using [artificial neural networks](#).^[137] Deepfakes have garnered widespread attention and concerns for their uses in [deepfake celebrity pornographic videos](#), [revenge porn](#), [fake news](#), [hoaxes](#), health [disinformation](#), [financial fraud](#), and covert [foreign election interference](#).^{[138][139][140][141][142]}

In July 2023, the fact-checking company [Logically](#) found that the popular generative AI models [Midjourney](#), [DALL-E 2](#) and [Stable Diffusion](#) would produce plausible disinformation images when prompted to do so, such as images of [electoral fraud](#) in the United States and Muslim women supporting India's [Hindu nationalist Bharatiya Janata Party](#).^[143]

Audio deepfakes

Instances of users abusing software to generate controversial statements in the vocal style of celebrities, public officials, and other famous individuals have raised ethical concerns over voice generation AI.^{[144][145][146][147][148][149]} In response, companies such as ElevenLabs have stated that they would work on mitigating potential abuse through safeguards and [identity verification](#).^[150]

Concerns and fandoms have spawned from [AI-generated music](#). The same software used to clone voices has been used on famous musicians' voices to create songs that mimic their voices, gaining both tremendous popularity and criticism.^{[151][152][153]} Similar techniques have also been used to create improved quality or full-length versions of songs that have been leaked or have yet to be released.^[154]

Generative AI has also been used to create new digital artist personalities, with some of these receiving enough attention to receive record deals at major labels.^[citation needed] The developers of these virtual artists have also faced their fair share of criticism for their personified programs, including backlash for "dehumanizing" an artform, and also creating artists which create unrealistic or immoral appeals to their audiences.^[citation needed]

Illegal imagery

Many websites that allow [explicit AI generated images or videos](#) have been created,^[155] and this has been used to create illegal content, such as [rape](#), [child sexual abuse material](#),^{[156][157]} [necrophilia](#), and [zoophilia](#).

Cybercrime

Generative AI's ability to create realistic fake content has been exploited in numerous types of cybercrime, including [phishing](#) scams.^[158] Deepfake video and audio have been used to create disinformation and fraud. In 2020, former Google [click fraud](#) czar [Shuman Ghosemajumder](#) argued that once deepfake videos become perfectly realistic, they would stop appearing remarkable to viewers, potentially leading to uncritical acceptance of false information.^[159] Additionally, [large language models](#) and other forms of text-generation AI have been used to create fake reviews of [e-commerce](#) websites to boost ratings.^[160] Cybercriminals have created large language models focused on fraud, including WormGPT and FraudGPT.^[161]

A 2023 study showed that generative AI can be vulnerable to jailbreaks, [reverse psychology](#) and [prompt injection](#) attacks, enabling attackers to obtain help with harmful requests, such as for crafting [social engineering](#) and [phishing attacks](#).^[162] Additionally, other researchers have demonstrated that open-source models can be [fine-tuned](#) to remove their safety restrictions at low cost.^[163]

Information laundering

Generative AI has been noted for its use by [state-sponsored propaganda campaigns](#) in [information laundering](#). According to a 2025 report by [Graphika](#), generative AI is used to launder articles from Chinese [state media](#) such as [China Global Television Network](#) through various social media sites in an attempt to disguise the articles' origin.^[164]

Reliance on industry giants

Training [frontier AI models](#) requires an enormous amount of computing power. Usually only [Big Tech](#) companies have the financial resources to make such investments. Smaller start-ups such as [Cohere](#) and [OpenAI](#) end up buying access to [data centers](#) from [Google](#) and [Microsoft](#) respectively.^[165]

Energy and environment

AI has a significant carbon footprint due to growing energy consumption from both training and usage.^[119] Scientists and journalists have expressed concerns about the environmental impact that the development and deployment of generative models are having: high CO₂ emissions,^{[166][167][168]} large amounts of freshwater used for data centers,^{[169][170]} and high amounts of electricity usage.^{[167][171][172]} There is also concern that these impacts may increase as these models are incorporated into widely used search engines such as Google Search and Bing,^[171] as [chatbots](#) and other applications become more popular,^{[170][171]} and as models need to be retrained.^[171]

The carbon footprint of generative AI globally is estimated to be growing steadily, with potential annual emissions ranging from 18.21 to 245.94 million tons of CO₂ by 2035,^[173] with the highest estimates for 2035 nearing the impact of the United States [beef industry](#) on emissions (currently estimated to emit 257.5 million tons annually as of 2024).^[174]

Proposed mitigation strategies include factoring potential environmental costs prior to model development or data collection,^[166] increasing efficiency of data centers to reduce electricity/energy usage,^{[168][171][172]} building more efficient [machine learning models](#),^{[167][169][170]} minimizing the number of times that models need to be retrained,^[168] developing a government-directed framework for auditing the environmental impact of these models,^{[168][169]} regulating for transparency of these models,^[168] regulating their energy and water usage,^[169] encouraging researchers to publish data on their models' carbon footprint,^{[168][171]} and increasing the number of subject matter experts who understand both machine learning and climate science.^[168]

Content quality

[The New York Times](#) defines [slop](#) as analogous to [spam](#): "shoddy or unwanted A.I. content in social media, art, books, and ... in search results."^[175] Journalists have expressed concerns about the scale of low-quality generated content with respect to social media content moderation,^[176] the monetary incentives from social media companies to spread such content,^{[176][177]} false political

messaging,[177]spamming of scientific research paper submissions,[178] increased time and effort to find higher quality or desired content on the Internet,[179] the indexing of generated content by search engines,[180] and on journalism itself.[181]

A paper published by researchers at Amazon Web Services AI Labs found that over 57% of sentences from a sample of over 6 billion sentences from [Common Crawl](#), a snapshot of web pages, were [machine translated](#). Many of these automated translations were seen as lower quality, especially for sentences that were translated into at least three languages. Many lower-resource languages (ex. [Wolof](#), [Xhosa](#)) were translated across more languages than higher-resource languages (ex. English, French).[182][183]

In September 2024, [Robyn Speer](#), the author of [wordfreq](#), an open source database that calculated word frequencies based on text from the Internet, announced that she had stopped updating the data for several reasons: high costs for obtaining data from [Reddit](#) and [Twitter](#), excessive focus on generative AI compared to other methods in the [natural language processing](#) community, and that "generative AI has polluted the data".[184]

The adoption of generative AI tools led to an explosion of AI-generated content across multiple domains. A study from [University College London](#) estimated that in 2023, more than 60,000 scholarly articles—over 1% of all publications—were likely written with LLM assistance.[185][[unreliable source?](#)] According to [Stanford University's](#) Institute for Human-Centered AI, approximately 17.5% of newly published computer science papers and 16.9% of peer review text now incorporate content generated by LLMs.[186]

If AI-generated content is included in new data crawls from the Internet for additional training of AI models, defects in the resulting models may occur.[187] Training an AI model exclusively on the output of another AI model produces a lower-quality model. Repeating this process, where each new model is trained on the previous model's output, leads to progressive degradation and eventually results in a "[model collapse](#)" after multiple iterations.[188]

On the other side, [synthetic data](#) can be deployed to train machine learning models while preserving user privacy.[189] The approach is not limited to text generation; image generation has been employed to train computer vision models.[189]